Vedant Krishnakant Mokal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.79
D.A.V Public School, New Panvel	Class XII (Central Board of Secondary Education)	2023	95.00%
D.A.V Public School, New Panvel	Class X (Central Board of Secondary Education)	2021	95.20%

WORK EXPERIENCE

Data Scientist Intern -- Macromill Inc., Japan

Tools: Python, Ollama, Databricks, Scipy, Numpy, Pandas, Streamlit

- Developed a binary classification framework for modeling human decision-making using the 19-dimensional Schwartz Theory of Basic Human Values and sparse spin-encoded survey data.
- Designed a quadratic scoring model using Frobenius inner products in Sym(19) to capture both individual value biases and pairwise value interactions.
- Evaluated the model using stratified 5-fold cross-validation, achieving 82.32% accuracy and outperforming the unweighted baseline (67.99%); benchmarked LLM-based and NLI-based feature extraction pipelines.

PUBLICATIONS

Probing dipole and quadrupole anisotropy in Gamma-ray bursts from Swift dataset.<Arxiv link>

- Investigated large-scale cosmic isotropy and the Cosmological Principle by analyzing spatial distributions in the 20-year Swift GRBcatalog using HEALPix spherical harmonic decomposition.
- Formulated 500isotropic Monte Carlo simulations to test dipole and quadrupole moment amplitudes against the null hypoth esis, identifying initial raw signatures of 2.9 and 7.2.
- ConstructedanopenSourceexposuremapmodelingSwiftpointinghistory,rollangles,andinstrumentPartialCodingFractions (PCF) to correct for directional observation biases, reducing spatial anisotropy significance to < 1.

PROJECTS

Sport Complex Crowd Monitoring System using YOLO: <github link>

Tools: Python, YOLOv11, RT-DETR, Roboflow datasets, Streamlit, OpenCV.

- Developed and trained model for Badminton and Table tennis settings with an objective to serve as accurate crowd monitoring system.
- Designed an interactive **Streamlit GUI** for live video analysis, model selection, and confidence tuning, integrated with OpenCV for visualization.
- Used **YOLOv11** object detection and tracking (ByteTrack, BoTSort) to detect the people and differentiate them on the basis of whether they are occupying the court or just bystanders.

Edge Plasma Turbulence Simulation using Hasegawa-Wakatani Models.<github link>

Tools: Python, Numpy, Cupy, Scipy, Scikit

- Developed a 2D pseudo-spectral solver for the ordinary and modified Hasegawa-Wakatani equations to simulate drift-wave turbulence in edge plasmas using FFTs and fourth-order Runge-Kutta integration.
- Implemented GPU acceleration using CuPy and benchmarked it against NumPy-based CPU execution for 2D FFT operations, enabling high-resolution 256 × 256 long-time plasma simulations with O(N2 logN) computational scaling.

Galaxy Morphology, Spin Direction, and Cosmic Anisotropy :<github link>

Tools:Python, Scikit-learn, TensorFlow/PyTorch, OpenCV, Emcee.

- Developed and benchmarked a suite of AI models (Hybrid Decision Tree, CNNs) to classify galaxy morphology and spin direction from astronomical images.
- Implemented a novel hybrid model, enhancing a deterministic analysis tool by feeding its output features into a Scikit-learn Decision Tree Regressor to improve predictive power.
- Applied Bayesian analysis with Emcee and frequentist methods to rigorously test cosmic anisotropy hypotheses and verify model findings against physical models.
- Developed an end-to-end Computer Vision model using deep learning (CNNs) for direct, image-based classification to create a fully autonomous AI pipeline.

RESEARCH EXPERIENCE

Working as an undergraduate researcher under Dr. Shantanu Desai

Tools: Python, Numpy,Scipy,Scikit,Emcee, Tensorflow, Corner.

- Developed Model analysis pipeline to perform scalable model fitting on large dataset using MCMC techniques.
- Applied Bayesian inference to estimate physical parameters which can be used in real world data analysis tasks.
- Built and Visualized probabilistic models, laying foundations for advanced ML workflow.
- Used PARAM SEVA supercomputer to parallelize the bayesian parameter estimation model using multiprocessing nodes and SLURM.

Affiliated Project: Using SFR-Stellar Mass Relation for Distance Measurement of Galaxies from JWST:<github link>

- Developed a Python-based framework to analyze premier JWST galaxy data , using the Star Formation Rate-Stellar Mass relation to measure luminosity distance and infer core cosmological parameters.
- Employed advanced Bayesian statistical analysis using the Emcee module to derive robust parametric relations from high-redshift galaxy datasets ($1.8 < z < 14$).
- Implemented data filtering and regression techniques to establish initial parameter estimates before applying sophisticated likelihood functions for robust parameter inference across multiple redshift and UV magnitude bins.

SKILLS

Tools and Frameworks: Git, Github, Anaconda, Django, Jupyter, MySQL, LaTeX, Solid Edge, Fusion360, Ansys OpticStudio, LTspice, SAOimageDS9

Libraries : Emcee(Bayesian analysis module) , OpenCV,Scipy, NumPy , Matplotlib, Astropy, SciKit, MySQL, GWcosmo,pandas,multiprocessing, tensor flow, openCV.

Techniques: Deep Learning (CNNs for Computer Vision), Decision Trees , Bayesian Analysis (Emcee), Statistical Modeling, Parameter Inference, Linear Regression , Data Filtering

CAD Software : Intermediate knowledge of SolidEdge and Fusion 360.

POSITIONS OF RESPONSIBILITY

Overall Head Cepheid :- The Astronomy Club of IITH

- Directed all club operations, managing resources such as telescopes and budgets.
- Spearheading the talent development at the club through collaborative projects and observational training with club equipment.
- leading a core team to plan and execute a diverse calendar of events including stargazing sessions, expert guest lectures, and hands-on workshops for the campus community.

Core Inbound and Media Cepheid :- Astronomy Club of IITH

- Led designing and implementing a custom automation system for the club telescope to enable advanced astrophotography.
- Developed and delivered workshops on AI/ML in astronomy, training members in ML techniques using Python.
- Organized and led numerous public outreach initiatives, including stargazing sessions and educational talks, to promote scientific literacy on campus.

Core GESTURE : The Arts and Crafts club of IITH

- Organized a diverse range of art and craft competitions, significantly increasing student participation and club visibility on campus.
- Facilitated regular, hands-on art sessions and workshops, fostering a collaborative community for creative expression and skill development.

WORKSHOPS AND PROFESSIONAL DEVELOPMENT

AI Winter School 2025-CFPU/Brown University Department of Physics

- Trained in applying advanced deep learning models for analyzing complex datasets in cosmology and physics.
- Mastered Bayesian inference and simulation-based techniques for robust statistical modeling.

ICTS Summer School on Gravitational Wave Astronomy

- Gained expertise in time-series analysis and advanced signal processing for extracting data from noisy environments.
- Applied Bayesian methods for parameter estimation of gravitational wave events, a high-stakes data analysis challenge.

RELEVANT COURSES

Computer Science and AI : Data Structures, Introduction to Programming, Introduction to AI

Mathematics: Introduction to Probability, Introduction to statistics, Elementary Linear Algebra, Differential equations, Number Theory, Calculus 1, Calculus 2, Maths for Physics, Advanced Math for Physics

Electronics: Basic Electrical Engineering, Basic Electrical Circuits, Digital Circuits, Analog Electronics

Languages: Spoken Japanese Basics, Introduction to German Language Level 1